

Xiao Sui

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RESEARCH INTERESTS

My research focuses on secure and scalable distributed computation, spanning Byzantine fault tolerance, blockchain systems, multiparty computation, and applied cryptography.

EDUCATION AND ACADEMIC POSITIONS

Postdoctoral Fellow (Shuimu Scholar) – the Institute for Advanced Study, Tsinghua University,
From August 2024 to present

Ph.D. in Cyberspace Security – Shandong University, June 2024

B.A. in Mathematics – Shandong University, June 2019

SELECTED PUBLICATIONS

X. Sui, S. Duan,

"Signature-Free Atomic Broadcast with Optimal $O(n^2)$ Messages and $O(1)$ Expected Time",
IEEE Symposium on Security and Privacy (S&P), 2025.

X. Meng, **X. Sui***, Z. Yang*, K. Rong, W. Xu*, S. Chen, Y. Yan, S. Duan*,

"Rondo: Scalable and Reconfiguration-Friendly Randomness Beacon",
Network and Distributed System Security (NDSS) Symposium, 2025.

S. Duan, H. Zhang, **X. Sui**, B. Huang, C. Mu, G. Di, X. Wang,

"Dashing and Star: Byzantine Fault Tolerance with Weak Certificates",
Proceedings of the Nineteenth European Conference on Computer Systems (EuroSys), 2024.

X. Sui, Q. Liu, S. Duan,

"Pike: Two-Phase BFT with Linearity and Flexible View Change",
IEEE Transactions on Computers (TC), 2025.

X. Sui, S. Duan, H. Zhang,

"BG: A Modular Treatment of BFT Consensus Towards a Unified Theory of BFT Replication",
IEEE Transactions on Information Forensics and Security (TIFS), 2023.

X. Sui, S. Duan, H. Zhang,

"Marlin: Two-Phase BFT with Linearity"

IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2022.

Honors & Awards

- **First Prize, Financial Technology Development Award**, People's Bank of China, 2024
(Recognized for contributions to blockchain consensus protocols and security; **the only student recipient** among the awardees)
- **Shuimu Scholar Postdoctoral Fellowship**, Tsinghua University, 2024
- **Outstanding Doctoral Dissertation Incentive Program**, CCF Blockchain Committee, 2025
- **China Postdoctoral Science Foundation (General Program)**, 2025

Academic Service & Professional Activities

- **Program Committee Member**,
ACM Conference on Computer and Communications Security (CCS), 2025-2026,
Blockchain and Distributed Systems Track
- **Reviewer** for IEEE Transactions on Information Forensics and Security (TIFS), Designs, Codes and Cryptography (DCC), and conferences/journals in security and cryptography.
- **Invited Talks**,
 - Huawei Sheldon Starry Sky Forum,
"BFT in Blockchain", 2024
 - CRYPTAPE RESEARCH,
"BFT in Blockchain: Optimizations and Applications", 2023
- **Contributed to the development of national and industry standards** related to blockchain consensus mechanisms and distributed systems security